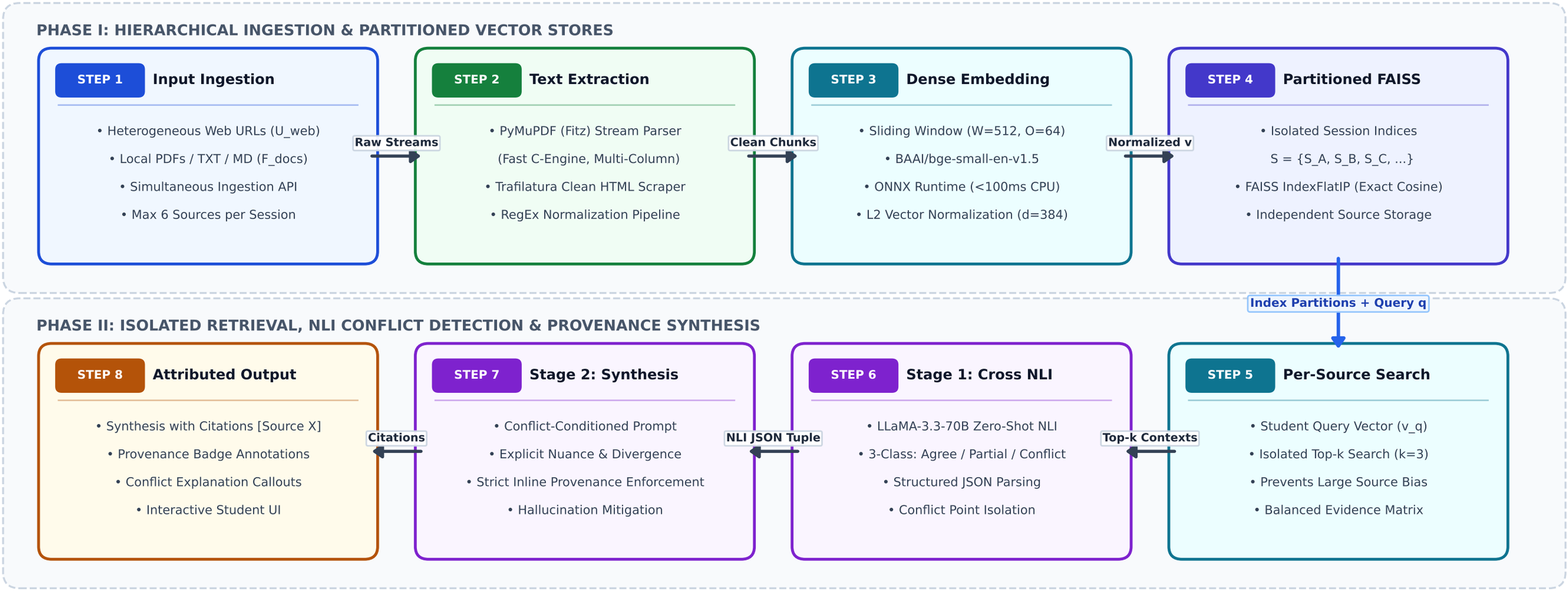
PHASE I: HIERARCHICAL INGESTION & PARTITIONED VECTOR STORES
STEP 1
Input Ingestion
• Heterogeneous Web URLs (U_web)
• Local PDFs / TXT / MD (F_docs)
• Simultaneous Ingestion API
• Max 6 Sources per Session
Raw Streams
STEP 2
Text Extraction
• PyMuPDF (Fitz) Stream Parser
(Fast C-Engine, Multi-Column)
• Trafilatura Clean HTML Scraper
• RegEx Normalization Pipeline
Clean Chunks
STEP 3
Dense Embedding
• Sliding Window (W=512, O=64)
• BAAI/bge-small-en-v1.5
• ONNX Runtime (<100ms CPU)
• L2 Vector Normalization (d=384)
Normalized v
STEP 4
Partitioned FAISS
• Isolated Session Indices
S = {S_A, S_B, S_C, ...}
• FAISS IndexFlatIP (Exact Cosine)
• Independent Source Storage
Index Partitions + Query q
PHASE II: ISOLATED RETRIEVAL, NLI CONFLICT DETECTION & PROVENANCE SYNTHESIS
STEP 8
Attributed Output
• Synthesis with Citations [Source X]
• Provenance Badge Annotations
• Conflict Explanation Callouts
• Interactive Student UI
Citations
STEP 7
Stage 2: Synthesis
• Conflict-Conditioned Prompt
• Explicit Nuance & Divergence
• Strict Inline Provenance Enforcement
• Hallucination Mitigation
NLI JSON Tuple
STEP 6
Stage 1: Cross NLI
• LLaMA-3.3-70B Zero-Shot NLI
• 3-Class: Agree / Partial / Conflict
• Structured JSON Parsing
• Conflict Point Isolation
Top-k Contexts
STEP 5
Per-Source Search
• Student Query Vector (v_q)
• Isolated Top-k Search (k=3)
• Prevents Large Source Bias
• Balanced Evidence Matrix